

LAB	PRACTICALS
18	Practice Lab to implement Aggregate Functions using GROUP BY and GROUP BY Having clauses. Create the following table, insert the given data and solve the following queries: - Total number of projects completed by each Team <pre>SELECT Team, SUM(ProjectsCompleted) AS TotalProjects FROM Employees GROUP BY Team;</pre> - Average years of service and maximum annual bonus within each Team <pre>SELECT Team, AVG(YearsService) AS AvgYearsService, MAX(AnnualBonus) AS MaxAnnualBonus FROM Employees GROUP BY Team;</pre> - Number of employees belonging to each Team <pre>SELECT Team, COUNT(*) AS TotalEmployees FROM Employees GROUP BY Team;</pre> - Total AnnualBonus grouped by both Team and YearsService <pre>SELECT Team, YearsService, SUM(AnnualBonus) AS TotalAnnualBonus FROM Employees GROUP BY Team, YearsService;</pre> - Teams that have more than two employees <pre>SELECT Team FROM Employees GROUP BY Team HAVING COUNT(*) > 2;</pre> - Teams whose employees have completed a combined total of more than 20 projects <pre>SELECT Team FROM Employees GROUP BY Team HAVING SUM(ProjectsCompleted) > 20;</pre> - Teams where the average annual bonus is less than 3500.00 <pre>SELECT Team FROM Employees</pre>

```
GROUP BY Team
```

```
HAVING AVG(AnnualBonus) < 3500.00;
```

8. Teams where the minimum number of projects completed by any single member is less than 5

```
SELECT Team
```

```
FROM Employees
```

```
GROUP BY Team
```

```
HAVING MIN(ProjectsCompleted) < 5;
```

9. Teams that have at least one employee with more than 8 years of service

```
SELECT DISTINCT Team
```

```
FROM Employees
```

```
WHERE YearsService > 8;
```

OR

```
SELECT
```

```
Team,
```

```
MAX(YearsOfService) AS TotalYearsService
```

```
FROM Employee
```

```
GROUP BY Team
```

```
HAVING MAX(YearsOfService) > 8;
```

10. Teams where the average annual bonus is greater than 3000.00
Order by average bonus in descending order

```
SELECT
```

```
Team,
```

```
AVG(AnnualBonus) AS AvgAnnualBonus
```

```
FROM Employees
```

```
GROUP BY Team
```

```
HAVING AVG(AnnualBonus) > 3000.00
```

```
ORDER BY AvgAnnualBonus DESC;
```

11. Teams where the sum of all members years of service is less than 15 years

```
SELECT Team
```

```
FROM Employees
```

```
GROUP BY Team
```

```
HAVING SUM(YearsService) < 15;
```

12. Find Teams where the maximum years of service is greater than 5
AND the minimum annual bonus awarded in that team is less than 4000.00.

```
SELECT Team
```

```
FROM Employees
```

```
GROUP BY Team
```

```
HAVING MAX(YearsService) > 5
```

```
AND MIN(AnnualBonus) < 4000.00;
```

19	Practice LAB to implement SELECT, UPDATE, ALTER, RENAME, TRUNCATE, DELETE, DROP, the LIKE Operator. Create the following table, insert the given data and solve the following queries: - Display all the records of DU_Students table SELECT * FROM DU_Students; - Display students studying in BCY-3 SELECT * FROM DU_Students WHERE Division = 'BCY-3'; - Display 5th semester students SELECT * FROM DU_Students WHERE Sem = 5; - Display contact number of all students SELECT ContactNo FROM DU_Students; - Display name and division SELECT Name, Division FROM DU_Students; - Extract last 3 digits of ContactNo SELECT RIGHT(ContactNo, 3) AS Last3Digits FROM DU_Students; - Display students sorted by Name (ascending) SELECT * FROM DU_Students ORDER BY Name ASC; - Display Name of Student who belongs to either Semester 3 or 5. (Use IN & OR) SELECT Name FROM DU_Students WHERE Sem IN (3, 5); SELECT Name FROM DU_Students WHERE Sem = 3 OR Sem = 5; - Find Student Name & Enrollment No in which Student Id between 103 to 105 SELECT Name, EnrollmentNo FROM DU_Students WHERE StuID BETWEEN 103 AND 105; - Find Student Name & Enrollment No with their Email Who belongs to 5th Semester. SELECT Name, EnrollmentNo, Email FROM DU_Students WHERE Sem = 5; - Display All the Details of first three students. SELECT TOP 3 * FROM DU_Students; - Display length of each student name SELECT Name, LEN(Name) AS NameLength FROM DU_Students;
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13. Update semester to 5 and Contact number to 98984477 whose id is 101.
`UPDATE DU_Students
SET Sem = 5, ContactNo = '98984477'
WHERE StuID = 101;`
14. Display Name & Enrollment no of first 30% Students
`SELECT TOP 30 PERCENT Name, EnrollmentNo FROM DU_Students;`
15. Display current date with student name.
`SELECT Name, GETDATE() AS CurrentDate FROM DU_Students;`
16. Display Unique Semesters.
`SELECT DISTINCT Sem FROM DU_Students;`
17. Display semester 5 student's division and Enrollment No
`SELECT Division, EnrollmentNo
FROM DU_Students
WHERE Sem = 5;`
18. Retrieve All the Students who have no Enrollment No
`SELECT * FROM DU_Students WHERE EnrollmentNo IS NULL;`
19. Display students whose email ends with 'gmail.com'
LIKE Operator
`SELECT * FROM DU_Students WHERE Email LIKE '%gmail.com';`
21. Modify ContactNo data type to VARCHAR(15)
`ALTER TABLE DU_Students
ALTER COLUMN ContactNo VARCHAR(15);`
22. Display students whose name contains 'Patel'
`SELECT * FROM DU_Students WHERE Name LIKE '%Patel%';`
23. Display student name in uppercase.
`SELECT UPPER(Name) AS StudentName FROM DU_Students;`
24. Rename column Name to StudentName
`EXEC sp_rename 'DU_Students.Name', 'StudentName';`
25. Update Division BCX-3 to BCX-4
`UPDATE DU_Students
SET Division = 'BCX-4'
WHERE Division = 'BCX-3';`
26. Update Division to BCX-5 & Semester to 5 whose Student Id Is 102
`UPDATE DU_Students
SET Division = 'BCX-5', Sem = 5`

`WHERE StuID = 102;`

27. Add one more Column City Varchar (50) in DU_Students table

`ALTER TABLE DU_Students`

`ADD City VARCHAR(50);`

28. Remove All BCX-4 Division Students

`DELETE FROM DU_Students`

`WHERE Division = 'BCX-4';`

29. Truncate the table (removes all data)

`TRUNCATE TABLE DU_Students;`

30. Remove table DU_Students from the database

`DROP TABLE DU_Students;`